MILITARY SPECIFICATION

DETECTOR, BATCH INTERFACE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a portable electronic batch interface detector.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/9	- Aluminum Alloy Bar, Rod, Shapes, Tube, and Wire, Extruded, 6063.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
VV-F-800	- Fuel Oil, Diesel.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-T-60	- Tape: Packaging, Waterproof.
PPP-T-75	- Tape, Preservation and Sealing.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Military

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

FSC 6680

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MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-S-3950	- Switches, Toggle, Environmentally Sealed, General Specification for.
MIL-C-5015	- Connectors, Electric, AN Type, General Specification for.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-S-8805	- Switches and Switch Assemblies, Sensitive, and Push (Snap Action), General Specification for.
MIL-C-13777/1	- Cable, Special Purpose, Electrical 2, 3, 4, and 5 Conductors.
MIL-C-13777/2	- Cable, Special Purpose, Electrical 6, 7, 8, 9, 10, 12, 13, and 15 Conductors.
MIL-C-13909	- Conduit, Metal, Flexible; Electrical, Shielded.
MIL-C-26482	- Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting.
MIL-C-39029/9	- Contacts, Electric, Pin, Crimp Removable, Composition C, Thermocouple.
MIL-C-39029/10	- Contacts, Electric, Socket, Crimp Removable, Composition C, Thermocouple.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment, Subsystem and System.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-1130	- Connections, Electrical, Solderless Wrapped.

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MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS3100	- Connector, Receptacle, Electric, Wall Mounting, Solder Contacts, AN Type.
MS3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MS3120	- Connectors, Receptacle, Electric, Crimp-Type, Wall Mounting, Flange, No. 4 Holes, Bayonet Coupling.
MS3121	- Connectors, Plug, Electric, Crimp Type, Cable-Connecting, Bayonet Coupling.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1298 - Density, Specific Gravity, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method, Test for.
E 100 - ASTM Hydrometers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA-No. 70).

National Electrical Code (Vol. 5, National Fire Codes).

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The batch interface detector (hereinafter called "detector") consists of the following components:

(a) Detector head, to be mounted on the pipeline.
(b) Thermocouple probe, to be mounted adjacent to the detector head.
(c) Signal and power cables.
(d) Receiver-converter.
(e) Indicator/recorder with alarm circuitry.
(f) Digital temperature indicator.
(g) Switches and wiring.

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(h) Weathertight instrument cabinet, equipped with removable, telescoping legs, and containing all instrumentation.
(i) Electric alarm bell and alarm light mounted on the outside of the instrument cabinet.

The detector head and receiver-converter operate together to detect change in the fuel specific gravity in a multiproduct petroleum pipeline.

3.1.1 Operational theory. The detector measures changes in fuel API gravity by measuring the effect of fuel API gravity on the frequency of vibration of a probe. A fixed-frequency vibration of the probe, which is totally immersed in the fuel stream, is initiated by a drive coil located in the detector head and powered by an oscillator in the receiver-converter. The fluid surrounding the probe dampens the frequency of vibration proportionally to the API gravity of the fuel. A pickup coil detects the dampened frequency and the receiver-converter converts this frequency shift to an output signal.

3.2 First article (preproduction model). The supplier shall furnish one or more detectors as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce detectors that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the supplier shall furnish to the Government one or more detectors for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4.1 Aluminum. Aluminum sheets and plates shall conform to QQ-A-250/11, Temper T4. Structural and special shapes shall conform to QQ-A-200/9, Temper T5.

3.5 Environmental requirements.

3.5.1 Operational. When unprotected by a shelter, the detector shall operate as specified herein in any ambient temperature from minus 25° F

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to plus 125° F. When the instrument cabinet is protected by a shelter, the detector shall operate as specified herein in any ambient temperature from minus 50° F to plus 125° F.

3.5.2 Standby. The detector shall not be damaged when in an operational standby posture, unprotected by a shelter, in air temperatures from minus 25° F to plus 160° F. When the instrument cabinet is protected by a shelter, the detector shall be capable of being maintained on standby status, undamaged, in temperatures from minus 50° F to plus 160° F.

3.6 Safety. The electronic and electrical elements of the detector shall meet the personnel safety requirements of MIL-STD-454, Requirement 1. The detector head and adjacent junction box shall be explosionproof, as defined in Article 501 of the National Electrical Code, for use in Class I, Division 1 locations.

3.7 Human factors. The detectors shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.8 Reliability. The specified mean-time-between-failure (MTBF) shall be 596 hours when the detector is tested as specified in 4.6.2.3 and 4.6.2.4.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.5 and 6.6), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.3 and 4.6.2.4.

3.9.2 Scheduled maintenance. The detector shall not require scheduled maintenance more often than that specified in the supplier's maintenance manual, furnished to the Government prior to any testing, when tested as specified in 4.6.2.8.2.

3.10 Ease of maintenance. All components shall be accessible, using common tools, for maintenance, repair, replacement, and adjustment without the removal of the other components, where possible, or with minimal disturbance of other components in cases where removal is necessary. The receiver-converter, recorder, and temperature indicator shall be equipped with slide-out racks or drawers.

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3.10.1 Plug-in connectors and modules. Plug-in connectors and modules shall be utilized whenever practicable. Plug-in modules (including printed circuit boards) shall be constructed so that physically interchanging any module with any other module is impossible.

3.11 Electric power requirements. The detector shall operate as specified herein on nominal 115 volts a.c., single phase, 60 Hertz (Hz). The detector shall exhibit no deterioration of performance when the nominal voltage varies as much as plus or minus 10 percent or when the frequency varies as much as plus or minus 5 percent.

3.12 Instrument cabinet.

3.12.1 Dimensions. The overall dimensions of the instrument cabinet, including visual and audible alarms but not including removable legs, shall be not more than 33 inches in length, 17 inches in width and 26 inches in height. The legs shall be removable and telescoping as to be adjustable over the range 24 to 48 inches. Connecting cables shall be as specified in 3.20.

3.12.2 Weight. The total weight of the instrument cabinet, including handles, lifting eyes, removable legs, and all the components fully assembled and installed, shall not exceed 150 pounds.

3.12.3 Design and fabrication. The instrument cabinet shall consist of an enclosure on a tripod base. The cabinet shall accommodate all of the instrumentation except the detector head and the thermocouple probe, and shall be fabricated of aluminum. The visual and audible alarms shall be mounted on top of the cabinet in weatherproof enclosures. The cabinet shall be provided with two completely weathertight, full-opening access doors, one in the front and one in the rear, equipped with latches that can be operated by personnel wearing arctic mittens. The doors shall provide full access to personnel for operation and maintenance of the detector. No venting shall be necessary, but drain holes shall be provided on the bottom. Instruments shall be mounted on a panel directly behind the front access door. Suitable cut-out portions shall be provided for the receiver-converter, recorder, temperature indicator, bell-silent button, night-light, and main switch.

3.13 Main switch. The main switch shall control all components of the detector. The switch shall be a double-pole single-throw (d.p.s.t.) switch of adequate capacity and shall conform to MIL-S-3950. It shall be mounted on the instrument panel, shall be labeled "MAIN SWITCH", and shall be self-indicating in the "ON" and "OFF" positions.

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3.14 Night-light. A night-light, with a standardized replaceable bulb of sufficient brightness to illuminate all controls and instruction plates, shall be installed on the instrument panel and shall be equipped with its own on-off switch.

3.15 Detector head and receiver-converter. The detector head and receiver-converter shall meet the following operational requirements:

(a) Operating range - 25 to 75 degrees API gravity (25° to 75° API) (0.90 to 0.68 specific gravity).
(b) Accuracy - within 0.1° API gravity only.
(c) Speed of response - nearly instantaneous.
(d) Output signal - 4 to 20 milliamps (mA), direct current (d.c.).

3.15.1 Detector head. The detector head shall consist of the sensing probe, the drive coil and the pickup coil, with the associated electrical connections, enclosed in a sealed metal housing. The metal housing shall be of noncorrosive metal and shall be able to withstand a pressure of 3000 pounds per square inch gage (psig). Auxiliary piping shall be Schedule 80 or equivalent. Zinc coating, cadmium plating, or use of brass, bronze or other copper-bearing alloys shall not be permitted on parts that are in contact with fuels. The detector head shall be mounted directly on the pipeline without bracing. The head shall fit a 1-1/2-inch NPT (female) connection. A drain valve shall be provided for the purpose of bleeding the detector head. The probe shall withstand complete immersion in the fuel and shall be capable of operating in an offset so that no portion of it need protrude inside the pipeline diameter. The detector head shall permit operation with not more than 18 inches of protrusion from the point of attachment on the pipeline. The probe shall not exceed 4 inches in length and shall be coated with polytetrafluoroethyl (TFE). A sheath or housing shall be provided to protect the probe from damage. The detector head shall include an explosionproof junction box of sufficient size to accommodate six terminal connectors with three 3/4-inch NPT ports. There shall be one port for the detector head, one for the thermocouple probe wires, and one for the conduit. Total weight of the detector head shall not exceed 20 pounds.

3.15.2 Receiver-converter. The receiver-converter shall operate in conjunction with the detector head. The receiver-converter shall be a single unit in a slide-out rack or drawer suitable for mounting in the instrument panel. The receiver-converter shall furnish a frequency-regulated power supply to the detector head and shall rectify and convert the input signal from the detector into a 4 mA through 20 mA output transmission to the recorder. In addition, the receiver-converter shall supply

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a zero suppression voltage to control the lower limit of the instrument span and a span voltage to control the upper limit of the instrument span. Controls shall be provided for ZERO and SPAN adjustment.

3.16 Recorder. The recorder shall be connected to, and record the output of, the receiver-converter. The recorder shall be a single unit installed in a slide-out rack or drawer mounted in the instrument panel. Recording shall be on 4-inch-strip chart paper with graduations from 25° to 75° API in linear horizontal divisions and 0 to 2400 hours in vertical time divisions. A scale for a pen position pointer shall be located above the strip chart and calibrated in the same gravity divisions as the chart paper. Two snap-type switches will be provided with adjustable position pointers on the same scale and covering the same range as the pen position pointer. The pen motion shall be linear to the input signal. The snap switches shall actuate an alarm whenever the pen position pointer deviates plus or minus 2° API. The recorder shall meet the following operational requirements:

(a) Input - 4 mA through 20 mA, d.c.
(b) Accuracy - within plus or minus 0.25 percent span.
(c) Threshold sensitivity - 0.17 mA.
(d) Speed of response - less than one second (full scale).
(e) Drift of pen position - less than 0.1 percent in 30 days.
(f) Ambient temperature effect - less than 0.5 percent due to a change of 50° F.

3.17 Alarms. The alarms shall consist of a visual and an audible alarm wired in parallel and actuated by the snap switches of the recorder. Electrical ratings shall be 115V a.c., 60 Hz, 5 amp.

3.17.1 Visual alarm. The visual alarm shall consist of an electric lamp assembly and a flasher unit. The electric lamp assembly shall include an electric lamp enclosed in a red globe protected by a metal shield. The lamp assembly shall be weatherproof and shall be mounted on top of the instrument cabinet. The flasher unit shall cause the lamp to flash at one- to three-second intervals. The flasher unit shall be a plug-in component, with a matching socket provided. When flashing, the visual alarm shall be visible for a distance of not less than 200 feet in all directions, with 2000-footlambert brightness, at any color between 1900 and 25,000 degrees kelvin.

3.17.2 Audible alarm. The audible alarm shall consist of an electric bell and a bell-silent switch. The electric bell shall be weatherproof and shall be mounted on top of the instrument cabinet. The bell-silent switch shall be wired in series with the bell to allow temporary silencing. The switch shall be a normally closed, single-pole single-throw (s.p.s.t.) momentary-contact pushbutton switch conforming to MIL-S-8805. The switch

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shall be mounted on the instrument panel and shall be labeled "BELL SILENCE". The audible alarm shall be audible for a distance of not less than 200 feet in all directions against background noise of not less than 60 decibels (based on a dB reference level of 0.0002 microbar).

3.18 Thermocouple probe and temperature indicator.

3.18.1 Thermocouple probe. The thermocouple probe element shall be of the copper/constantan type and shall be sensitive over the range of -50° F to +125° F. One copper lead and one constantan lead, not less than 5 feet in length, shall be included for the purpose of connecting the probe to the temperature indicator. The probe housing shall be not more than 6 inches long and a 1/4-inch NPT bushing shall be included for mounting. The thermocouple probe housing and bushing shall be of noncorrosive metal(s) and shall be capable of withstanding a pressure of 3000 psig.

3.18.2 Temperature indicator. The temperature indicator, operating in conjunction with the thermocouple probe, shall indicate the fuel temperature by means of a digital display over the range -50° F to +125° F with an accuracy of ± 2° F. The temperature indicator shall be a single unit in a slide-out rack or drawer mounted in the instrument panel.

3.19 Signal and power receptacles. Signal and power receptacles shall be mounted adjacently on the exterior of the instrument cabinet in a conspicuous, protected location.

3.19.1 Power receptacle. A three-contact pin connector receptacle, conforming to MIL-C-5015 and MS3100F16-10P, shall mate with the power-cable plug to provide the detector with 115V, 60 Hz power.

3.19.2 Signal receptacle. A signal receptacle conforming to MIL-C-26482 and MS3120F12-10P shall provide the input and output for both the detector head and the pair of thermocouple conductors in the signal cable. The two wires for the thermocouple pair shall consist of one copper wire and one constantan wire. The signal receptacle shall mate with the signal-cable plug by means of two-pin contacts which shall be copper/constantan couples conforming to MIL-C-39029/9-20-20-C1 (copper) and MIL-C-39029/9-20-20-C5 (constantan).

3.20 Connecting cables. Two connecting cables shall be supplied: a power cable to connect the instrument cabinet with the power supply and a signal cable to connect the detector head junction box with the instrument cabinet.

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3.20.1 Power cable. The power cable shall be a three-conductor cable 200 feet in length conforming to MIL-C-13777/1, Type 030565. A three-contact socket plug connector, conforming to MIL-C-5015 and MS3106F16-10S, shall be installed on one end of the cable for connection to the power receptacle on the instrument cabinet.

3.20.2 Signal cable. The signal cable shall be a six-conductor cable 200 feet in length conforming to MIL-C-13777/2, Type 060480. Two pairs shall be shielded and jacketed and the third pair shall be solid thermocouple copper-constantan. A 10-contact socket connector plug conforming to MIL-C-26482 and MS3121F12-10S shall be installed on one end. Two of the socket contacts shall be copper/constantan couples conforming to MIL-C-39029/10-20-20-C1 (copper) and MIL-C-39029/10-20-20-C5 (constantan). The copper and constantan conductors shall be fitted to the copper and constantan contacts. The socket connector plug shall mate with the 10-contact pin connector receptacle on the instrument cabinet. The copper socket shall mate with the copper pin and the constantan socket shall mate with the constantan pin. The other end of the cable shall have its conductors laid bare for insertion through the conduit to the detector head junction box at assembly.

3.21 Wiring and cabling.

3.21.1 Connections. Wires subject to breakage at the connections shall be provided with terminals that grip the wire insulation. No electrical connections shall be made by clamping a conductor between a metallic member and an insulating material of other than a ceramic or vitreous nature. Wire with outer braid covering (fabrics) shall have the ends of such braid secured against fraying by mechanical means or by application of varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.24). Solderless, wrapped connections shall be in accordance with MIL-STD-1130.

3.21.2 Plug-in connections. Plug-in connections shall be used wherever practicable. Screw terminals, soldered connections and solderless crimp-type connections shall be used, in that decreasing order of preference. Leads to those wires which are to be soldered shall be sufficiently spaced to prevent damage to nearby leads or parts when soldering the lead. Terminals or other connections to which wires are to be soldered shall be sufficiently long to prevent damage to insulation and surrounding materials by a hot soldering iron. Solderless connectors shall be used for connection of the detector head leads to the signal-cable leads. Wire nuts shall be used for connection of the thermocouple probe leads to the signal-cable thermocouple leads.

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3.21.3 Hookup wire selection, application and identification. Hookup wire selection, application and identification shall be in accordance with MIL-STD-454, Requirement 20.

3.21.4 Insulating sleeving. Insulating sleeving shall be in accordance with MIL-STD-454, Requirement 11.

3.21.5 Slack. Wires and cables shall be as short as practicable, except that sufficient slack shall be provided to:

(a) Prevent undue stress on cable forms, wires and connections.
(b) Enable parts to be removed and replaced during servicing without disconnecting other parts.
(c) Facilitate field repair of broken or cut wires.
(d) Permit pulling the units out of drawers or slide-out racks without breaking connections. (Units which are difficult to connect when mounted shall be capable of movement to a more convenient position for connecting and disconnecting cables.)

3.22 Accessories. The following accessories shall be furnished with each detector:

(a) Conduit, metal, rigid, zinc coated, nominal size 3/4 inch, 30 feet in 5-foot lengths with couplings, conforming to WW-C-581.
(b) Conduit, metal, flexible, nominal size 3/4 inch, 18 inches long, conforming to MIL-C-13909, Type II, Grade A.
(c) Recorder, ink, compatible with the recorder capable of flowing at any temperature between minus 50° F and plus 125° F.
(d) Chart paper to fit the recorder as specified in 3.16.

3.23 Electromagnetic interference. The electromagnetic interference emission characteristics of the detector shall conform to MIL-STD-461, Class IIA.

3.24 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.24.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

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(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with adherence or function of the varnish.

3.24.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of Composition I. When Composition II is used, the supplier shall provide evidence to the Government that the use of Composition II is required and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.25 Identification marking. The detector shall be identified in accordance with MIL-STD-130. The marking shall be applied to the detector on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts or rivets in a location where the plate will be both visible and legible.

3.26 Instruction plates. Each detector shall be equipped with one or more instruction plates describing special or important procedures to be followed in assembling, operating or servicing the detector. In addition, a plate shall be provided which bears the API Gravity Correction Chart shown in Figure 1. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts or rivets in a location where the plates will be both visible and legible.

3.27 Treatment and painting. The portions of the instrument cabinet and removable legs normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, Type A, Color No. 24087.

3.28 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

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3.29 Workmanship.

3.29.1 Castings and forgings. All parts, components and assemblies of the detector which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.29.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.29.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler materials, and fabrication techniques shall be as required to enable the detector to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and shall in no event be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the detector shall be at the option of the supplier.

3.29.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.29.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

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3.29.6 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein and to the standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards and as specified herein.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of preparation for delivery (see 4.8).

4.3 Preproduction inspection. Prior to the examination and testing of the detector, the following shall be performed:

(a) Have available the means to provide the detector with an operating voltage of 115V, 60 Hz.
(b) Place the detector head probe in a container so that it can be completely covered with fuel.
(c) Have on hand appropriate quantities of the following fuels sufficient to fill the container mentioned in (b) above:

(1) Turbine fuel - MIL-T-5624, Grade JP-5.
(2) Diesel oil - VV-F-800.
(3) Automotive gasoline - MIL-G-3056.

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(d) Have available hydrometers covering the range 25° - 75° API and conforming to ASTM E 100, Nos. 3H-62 - 8H-62, 26H-62 - 35H-62, 53H-62 - 58H-62, or equivalent.

(e) Have available a calibrated thermometer covering the range of minus 50° F to plus 150° F and accurate to 0.5° F.

4.3.1 Examination. The preproduction detector shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction detector shall be tested as specified in 4.6.2.1 through 4.6.2.10. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production detectors will be selected at random by the Government from the detectors being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2.1 through 4.6.2.10 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production detector shall not exclude the remaining detectors from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production detector to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production detector and shall be cause for refusal by the Government to continue acceptance of production detectors until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on detectors previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all detectors accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Examination. Each detector shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection of the detector.

4.5.2 Tests. Each detector shall be tested as specified in 4.6.2.2 and 4.6.2.3. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The detector shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Safety requirements not as specified.
103. Components not accessible for maintenance.
104. Slide-out racks and drawers not provided where required.
105. Plug-in modules interchangeable.
106. Electric power requirements not as specified.
107. Instrument cabinet not as specified.
108. Main switch not as specified.
109. Night-light not as specified.
110. Detector head not as specified.
111. Receiver-converter not as specified.
112. Recorder not as specified.
113. Visual alarm not as specified.
114. Audible alarm not as specified.
115. Thermocouple probe not as specified.
116. Temperature indicator not as specified.
117. Signal and power receptacles not as specified.
118. Connecting cables not as specified.
119. Wiring and cabling not as specified.
120. Accessories not as specified.
121. Fungus and moisture resistance not as specified.
122. Workmanship not as specified.

Minor

201. Marking not as specified.
202. Instruction plates not as specified.
203. Treatment and painting not as specified.

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4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The detector shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test.

4.6.2.2 Detection and recording. Turn on the detector and let it warm up for 5 minutes. Open the detector head valve and fill the test container with diesel fuel, making sure that the probe is completely immersed. Measure the API gravity with a hydrometer in accordance with ASTM D 1298; do not correct to 60° F. If the gravity indicated on the recorder is not that indicated by the hydrometer, adjust the ZERO control until they coincide. Adjust the alarm pointers on the recorder so that one pointer is set 2° API above, and the other pointer 2° API below the pen position pointer. Let stand for a minimum of 10 minutes. Open the test container drain valve and, while the diesel fuel is draining out, slowly pour in turbine fuel JP-5, making sure that the detector head probe is completely immersed at all times. Shut off drain when a minimum change of 7° API is indicated by the hydrometer. Note the gravity shown by the hydrometer and compare it to that indicated by the pen position pointer. If pen movement is insufficient, increase the SPAN setting on the receiver-converter by the appropriate amount (e.g., if pen movement is only 75 percent of actual change, increase SPAN setting by 25 percent). Similarly, if pen movement is too great, decrease SPAN setting by the appropriate amount. Adjust the ZERO control until the pen position pointer indicates the gravity indicated by the hydrometer. Adjust alarm pointers 2° API each side of the pen position pointers. Slowly drain out the turbine fuel and replace with diesel fuel until an observed change of slightly more than 2° API is indicated by the hydrometer. Observe the change in the pen position pointer. Push the BELL SILENCE switch to momentarily silence the audible alarm. If the gravity indicated by the recorder is not within 0.5° API of that indicated by the hydrometer, adjust the ZERO control until they coincide. Reset alarm indicators and repeat the test. Tests shall be repeated until two complete tests are performed without adjustment of the SPAN control. Any one of the following constitute failure of this test:

(a) Failure of the ZERO control to move the recorder pen.
(b) Failure of the SPAN control to change the amount of pen movement per change in API gravity.
(c) Failure of the detector to indicate proper gravity change, as measured by the hydrometer, to within ±1° API.

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(d) Failure of the detector to indicate a gravity change of at least 2° API when the gravity change of the fuel, as measured by the hydrometer, is 2° API.
(e) Failure of the alarm system to activate when a gravity change of 2° API is indicated.
(f) Failure of the BELL SILENCE switch to shut off audible alarm.

4.6.2.3 Stabilization and temperature indication. Turn on the detector and let it warm up for 5 minutes. Fill the test container with any petroleum fuel specified in 4.3. Open the detector head drain valve and immerse the detector head probe in the fuel. Also immerse the thermocouple probe up to its bushing. Using a calibrated thermometer, measure the fuel temperature to 0.5° F. Measure the observed API gravity with a hydrometer in accordance with ASTM D 1298; do not correct to 60° F. If the gravity indicated on the recorder is not that indicated by the hydrometer, adjust the ZERO control until they coincide. Adjust the alarm pointers on the recorder so that one pointer is set 2° API above, and the other pointer is set 2° API below the pen position pointer. Allow the detector to run for 24 hours at a constant temperature. Check temperature at start and end of the 24-hour period and examine the recorder chart after 24 hours. Remeasure the observed API gravity with a hydrometer in accordance with ASTM D 1298. Any one of the following shall constitute failure of this test:

(a) Failure of the temperature indicator to indicate correct temperature, as measured by the thermometer, to within 2° F.
(b) Failure of the recorder to ink properly.
(c) Movement of the pen more than 1° API in 24 hours if observed API gravity (hydrometer) has remained constant.

4.6.2.4 Low temperature. A tent shelter sufficient to provide wind protection shall be provided for the detector instrument cabinet. The entire detector with the shelter shall be placed in a cold chamber and the temperature lowered to the requirements of 3.5.1 or 3.5.2, as applicable, and allowed to remain until the temperature is stabilized. After stabilizing, the tests specified in 4.6.2.2 and 4.6.2.3 shall be performed, except that automotive gasoline may be substituted for diesel fuel. Failure as defined in 4.6.2.2 and 4.6.2.3 shall constitute failure of this test.

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4.6.2.5 High temperature. The detector shall be warmed to the temperature requirements of 3.5.1 or 3.5.2, as applicable. After warming, the tests specified in 4.6.2.2 and 4.6.2.3 shall be performed. Only diesel fuel and turbine fuel JP-5, may be used. Do not use any fuel that has a flash point of under 125° F. Failure as defined in 4.6.2.2 and 4.6.2.3 shall constitute failure of this test.

4.6.2.6 Human factors. The detector design shall be examined for compliance with the applicable requirements for visual display, scale indicator, plotters, auditory display, and labeling and design for maintainability in accordance with MIL-STD-1472. Nonconformance to the applicable requirements of MIL-STD-1472 shall constitute failure of this test.

4.6.2.7 Reliability. Using the MTBF specified in 3.8, the detector shall be tested as specified in 4.6.2.2 and 4.6.2.3 with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VI. Continue testing until either an "accept" or "reject" decision is reached. For the purpose of this test, a failure is as defined in 4.6.2.2 and 4.6.2.3.

4.6.2.8 Maintainability.

4.6.2.8.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the detector as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.2.8.2 Scheduled maintenance. Scheduled maintenance shall be performed during initial production testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.23. The supplier shall furnish to the contracting officer a report on the test required by MIL-STD-461 within the time frame specified (see 6.2). Disapproval of the report shall constitute failure of this test.

4.6.2.10 Pressure test. The detector head and thermocouple probe shall be installed in a pressure vessel and pressure-tested with water or fuel. Pressure shall be held at 3000 psig for 1 minute. Leakage of any kind shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select detectors at any time during the contract production period and subject these detectors to the examination specified in 4.6.1 and the tests specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on detectors selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production detectors. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison detector to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison detector and shall be cause for refusal by the Government to continue acceptance of production detectors until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on detectors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all detectors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery.

4.8.1 Preproduction pack inspection.

4.8.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.1.2 Tests. The preproduction pack of the boxed detector shall be subjected to one of the guided-impact tests specified in MIL-STD-1186, Appendix A.

4.8.1.2.1 Railroad-car test. When the railroad-car impact test is used, the car shall strike a string of five empty cars, with draft gear extended and the brake set, at a speed of not less than 10, nor greater than 11, miles per hour.

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4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

123.	Materials, methods and containers not as specified. Each incorrect material, method or container shall constitute one defect.
124.	Rigid conduit not bundled as specified for Level A.
125.	Cables not secured and bare ends not taped as specified for Level A.
126.	Cabinet legs not removed from cabinet for Level A.
127.	Receptacles and openings not covered as specified for Level A.
128.	Fiberboard boxes not closed as specified for Level A.
129.	Closure and strapping not as specified for Level A or B.
130.	Marking missing, illegible or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packaging, packing, and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be prepared utilizing either the preproduction model or a production detector. When the preproduction model is utilized, any preservation, packaging and packing shall be removed by the supplier at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production detectors.

5.2 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Unprotected surfaces. Unprotected metal surfaces requiring a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

5.2.1.2 Chart paper. The chart paper shall be preserved by Method IC-1 or IC-3 of MIL-P-116.

5.2.1.3 Ink. The ink shall be packaged in a box conforming to PPP-B-566, Variety 1.

5.2.1.4 Rigid conduit. The rigid conduit shall be bundled together. The bundle shall be wrapped with tape at two locations, equidistant from each other and from the ends of the conduit. The tape shall conform to PPP-T-97, Type I, and shall be not less than 3/4 inch wide.

5.2.1.5 Detector head and thermocouple probe. The detector head and thermocouple probe shall be packaged in a fiberboard box conforming to PPP-B-636, Class Weather Resistant. The box shall be closed by Method V of the appendix to the box specification.

5.2.1.6 Cables. The cables shall be coiled to allow packing in the shipping container. Each individual coil shall be tied or taped at three places spaced equidistantly along the circumference of the coil. Bare ends of conductors shall be wrapped with tape conforming to PPP-T-60, Type III or IV, applied over the ends of the wire and 4 inches back over the insulation.

5.2.1.7 Cabinet legs. The cabinet legs shall be removed from the instrument cabinet and packaged in a fiberboard box conforming to PPP-B-636, Grade W5c. The box shall be closed by Method V of the appendix to the box specification.

5.2.1.8 Receptacles and openings. All receptacles and openings in the equipment shall be covered with tape conforming to PPP-T-75, Type I or II.

5.2.2 Level C. The detector and accessories shall be packaged to afford protection against deterioration and damage during shipment from the supplier to the initial destination.

5.3 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

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5.3.1 Level A. The detector with its accessories shall be packed in a close-fitting box conforming to PPP-B-601, Overseas Type, Grade B, or PPP-B-621, Class 2. The contents shall be blocked, braced and cushioned in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.3.2 Level B. The detector and accessories shall be packed as specified in 5.3.1 for Level A, except the containers shall be Domestic Type.

5.3.3 Level C. The detector shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The detector is intended to be used under both normal and extreme climatic conditions for interface detection in all military overland multiproduct pipeline systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model and number of detectors required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required and number of detectors to be furnished (see 3.3).
(e) When the welding process used in fabrication is to be other than as specified (see 3.29.3).

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(f) Time frame required for submission of electromagnetic interference test report, when the preproduction model is so tested (see 4.6.2.9).
(g) Time frame required for submission of preproduction pack (see 5.1).
(h) Level of preservation and packaging and level of packing required (see 5.2 and 5.3).

6.3 Preproduction model. Any changes or deviations of production detectors from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish detectors conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, package, pack, and mark the detectors in accordance with this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each detector.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of detectors.

Custodian:

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Preparing activity:

Army - ME

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ACTUAL FUEL TEMP	DEGREES API CORRECTION				
	AV GAS	MO GAS	JP-4	JP-5	DIESEL
- 20° F	+ 9.0°	+ 7.2°	+ 8.6°	+ 6.0°	+ 5.0°
- 10°	+ 7.9°	+ 6.3°	+ 7.5°	+ 5.2°	+ 4.4°
0°	+ 6.7°	+ 5.4°	+ 6.4°	+ 4.5°	+ 3.8°
10°	+ 5.6°	+ 4.5°	+ 5.4°	+ 3.8°	+ 3.1°
20°	+ 4.5°	+ 3.6°	+ 4.3°	+ 3.0°	+ 2.5°
30°	+ 3.4°	+ 2.7°	+ 3.2°	+ 2.2°	+ 1.9°
40°	+ 2.2°	+ 1.8°	+ 2.1°	+ 1.5°	+ 1.3°
50°	+ 1.1°	+ 0.9°	+ 1.1°	+ 0.8°	+ 0 6°
60°	CORRECTIONS ABOVE, ADD TO PEN READING NO CORRECTION NECESSARY AT 60° CORRECTIONS BELOW, SUBTRACT FROM PEN READING				
70°	- 1.1°	- 0.9°	- 1.1°	- 0.8°	- 0.6°
80°	- 2.2°	- 1.8°	- 2.1°	- 1.5°	- 1.3°
90°	- 3.4°	- 2.7°	- 3.2°	- 2.2°	- 1.9°
100°	- 4.5°	- 3.6°	- 4.3°	- 3.0°	- 2.5°
110°	- 5.6°	- 4.5°	- 5.4°	- 3.8°	- 3.1°
120°	- 7.9°	- 5.4°	- 6.4°	- 4.5°	- 3.8°
	INTERPOLATE AS REQUIRED TO NEAREST 5°				

FIGURE I. API GRAVITY CORRECTION CHART

X-2651

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-D-52840(ME) Detector, Batch Interface

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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